

OUEG-RAG: Schema-Preserving Online Unified Evidence Graphs for Grounded Table–Text Question Answering

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Abstract. Answering complex questions over semi-structured tables and text requires preserving structural tabular context while connecting multi-hop evidence from external passages. Existing retrieval-augmented generation approaches face a fundamental dilemma: flattening tables into plain text obscures hierarchical header associations and intra-row bindings, whereas graph-based methods typically prune evidence into a few linear paths, inadvertently discarding vital peripheral facts and failing under structural candidate ties. We propose the Online Unified Evidence Graph framework for retrieval-augmented generation, constructing a bounded, schema-preserving multigraph dynamically at query time. The proposed framework combines a two-stage retriever with hyperlink-guided bridge passage expansion, preserving qualified headers and row bindings alongside deterministic text triples. Crucially, rather than applying destructive path filtering, the architecture couples complete evidence subgraph serialization with online grounded path traversal—leveraging dynamically enumerated multi-hop paths as non-filtering reading guides to steer language model attention without losing contextual evidence. Furthermore, the framework introduces an active structural ambiguity evaluator that detects candidate score collisions and supplies lexical tiebreak signals. Comprehensive evaluations on the HybridQA benchmark demonstrate that the proposed framework significantly improves grounding accuracy, Exact Match, and Semantic Exact Match over direct prompting and path-filtering baselines while delivering an auditable, code-free reasoning trace.

Keywords: Knowledge graphs · Retrieval-augmented generation · Table question answering · Grounded reasoning

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1 Introduction

Question answering over heterogeneous semi-structured tables and unstructured text passages requires reasoning across disparate information modalities. In the standard HybridQA benchmark [4], answering a multi-hop question requires locating an entity within a table cell, traversing its Wikipedia hyperlink to an external unstructured passage (or vice-versa), and extracting the grounded answer span. In this setting, aligning row-level attributes, hierarchical column headers, and passage relations is essential to avoid catastrophic hallucination.

The predominant baseline in retrieval-augmented generation (RAG) [12, 8, 7] is Direct LLM prompting (in-context generation), where serialized table rows and retrieved passages are supplied directly to the language model. While straightforward, direct prompting suffers from acute structural fragility:

1. *Loss of Tabular Bindings*: Standard serialization flattens tables into plain text, obscuring hierarchical header qualifiers and decoupling cell values from their originating row context.
2. *Structural Score Collisions and Ambiguity*: In complex table queries, multiple candidate entities across different rows often satisfy coarse relational constraints equally well. Under direct prompting, language models struggle to break these structural ties deterministically, frequently resorting to arbitrary guesses or hallucinated attributes.

To impart structure, previous graph-augmented methods construct knowledge graphs and extract candidate reasoning paths [21, 14, 16]. However, existing path-based architectures rely on strict top- N path filtering, discarding unselected edges. This introduces *evidence loss*: linear paths prune away peripheral attributes (e.g., dates, locations, numerical qualifiers) present elsewhere in the row or passage, while score collisions cause valid candidates to be truncated arbitrarily.

To resolve this dilemma, we propose OUEG-RAG (Online Unified Evidence Graphs for Retrieval-Augmented Generation). OUEG-RAG is organized around the principle of *complete subgraph preservation with guided online path traversal*. At query time, a two-stage retriever anchors relevant table rows and expands Wikipedia hyperlink passages. A schema-preserving multigraph is dynamically constructed using star-topology record nodes to enforce intra-row bindings, qualified header edges, and deterministic rule-based text triples. Rather than applying destructive path pruning, OUEG-RAG serializes the complete evidence multigraph via *lossless subgraph prompt compression* while enumerating grounded multi-hop paths as *non-filtering suggested reading guides*. Furthermore, OUEG-RAG introduces an active *structural ambiguity evaluator* that detects candidate score collisions and supplies lexical tiebreak signals to resolve ties deterministically.

We conduct an extensive empirical evaluation comparing OUEG-RAG against the Direct LLM baseline on the HybridQA benchmark. Rather than relying on rigid offline pipelines or heuristic prompt engineering, our empirical findings demonstrate three consistent advantages: (1) *Generalizable Grounding Gains*:

Structuring heterogeneous table cells and linked text into a schema-preserving multigraph consistently improves exact match and token-level F1 across different model scales, achieving substantial net positive margins in head-to-head pairwise evaluations. (2) *Decisive Resilience Under Structural Ambiguity*: The empirical advantage of OUEG-RAG is particularly pronounced on complex queries exhibiting structural score collisions across candidate table rows. On this subset, active score collision detection and lexical tiebreaking systematically avert the arbitrary guesses and hallucinations that plague direct prompting. (3) *Preservation of Context Integrity*: By bounding candidate table rows and external hyperlink passages within a compressed evidence multigraph, OUEG-RAG eliminates context-budget truncation errors while incurring a modest, predictable latency trade-off. A comprehensive quantitative breakdown across metrics, model configurations, ambiguity partitions, and runtime latencies is presented in Section 5 and Section 6 (Table 1).

We articulate four main methodological contributions:

- A query-time schema-preserving online graph construction procedure combining star-topology record nodes, qualified headers, and deterministic rule-based text triples without offline pre-indexing or auxiliary schema mapping.
- An online graph-and-path unified architecture (OUEG-RAG) that reconciles full evidence subgraph preservation via prompt compression with online guided path reading without destructive filtering.
- *Structural Ambiguity Assessment and Tiebreak (Ambiguity Assessment & Tiebreak)*: An active mechanism that automatically detects structural score-collision ties when multiple candidate entities satisfy relational constraints with identical graph scores, resolving ambiguities via bridge lexical keywords to prevent hallucination.
- *Empirical Verification Across Model Scales*: A comprehensive head-to-head evaluation across model scales, demonstrating consistent gains in Exact Match, Token F1, and resilience against context truncation over Direct LLM prompting on HybridQA.

2 Related Work

Retrieval-Augmented Generation (RAG) and Table-RAG. Retrieval-Augmented Generation [12, 8] augments LLMs with non-parametric knowledge retrieved via dense or lexical retrievers [11, 7]. However, conventional chunk-based RAG is tailored for flat passages; linearizing table rows into plain text destroys geometric coordinates, detaches cell values from column qualifiers, and triggers hallucinations when candidates share overlapping surface forms. Early table comprehension relied on semantic parsing over isolated tables [15] or table-specific pretraining [10, 22, 13]. Recent efforts scale LLM table reasoning via schema-and-cell retrieval [3] or SQL generation across documents [23]. In hybrid multi-hop settings like HybridQA [4], questions demand simultaneous traversal across table cells and hyperlinked passages. Unlike code interpreters or single-table re-

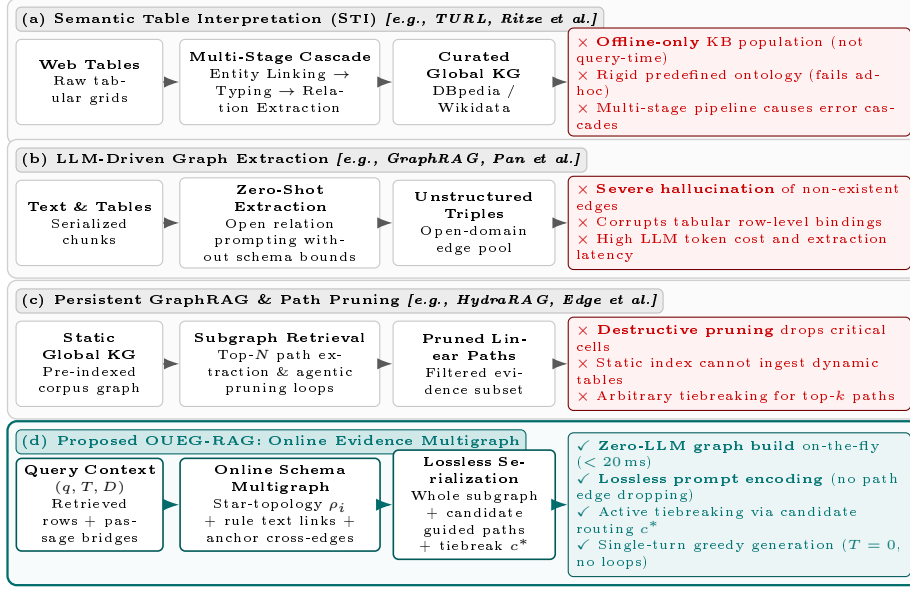


Fig. 1. Architectural comparison across four knowledge graph construction and reasoning paradigms for hybrid table-text retrieval. In contrast to offline STI cascades (a), hallucination-prone zero-shot LLM extraction (b), and destructive top- k path pruning in persistent GraphRAG systems (c), the proposed OUEG-RAG framework (d) dynamically instantiates a schema-preserving multigraph on-the-fly without LLM calls, achieves lossless subgraph prompt encoding preserving all retrieved context, and guides single-turn reasoning via candidate path routing.

trievers, OUEG-RAG constructs a lightweight, schema-preserving multigraph dynamically at query time without requiring SQL parsers or sandbox execution.

Knowledge Graph Paradigms: SOTA vs. OUEG-RAG. Extracting structured representations from tables and texts encompasses three prior paradigms alongside our proposed framework (Fig. 1):

1. *Semantic Table Interpretation (STI)*: Traditional pipelines [17, 5] convert web tables into formal knowledge graphs via multi-stage cascades (Entity Linking → Typing → Relation Extraction) aligned to curated ontologies (e.g., DBpedia, Wikidata; Fig. 1a). While standardized, STI requires predefined rigid ontologies, suffers from error cascades, and targets offline KB population rather than dynamic query-time reasoning.
2. *LLM-Driven Triplet Extraction*: Prompting LLMs to extract open relational triples from serialized chunks [6, 14] is schema-free, but suffers from severe hallucinations of phantom edges, high token latency, and destroys rigid tabular row-level bindings (Fig. 1b).
3. *Persistent GraphRAG & Path Pruning*: Graph RAG architectures [6, 20, 16] index an entire corpus into a pre-built static KG and retrieve subgraphs

at runtime. However, static graphs cannot ingest dynamic ad-hoc tables, and aggressive top- k pruning destructively discards peripheral cells while resolving score ties arbitrarily (Fig. 1c).

The OUEG-RAG Paradigm: As contrasted in Fig. 1(d), OUEG-RAG departs from existing paradigms across four fundamental dimensions: (1) *Ephemeral vs. persistent*: It builds an online multigraph in < 20 ms directly from query-associated evidence without pre-indexed KGs, preserving row bindings via star centroids ρ_i . (2) *Evidence preservation vs. destructive pruning*: Rather than discarding peripheral cells via pruning, it serializes the complete subgraph losslessly, employing scored candidate paths strictly as non-filtering reading guides. (3) *Structural tiebreaking*: When candidate rows share identical graph scores, OUEG-RAG resolves collisions deterministically (c^*) via bridge lexical overlap rather than arbitrary truncation. (4) *Zero-LLM graph build*: Graph construction uses deterministic rules, enabling single-turn greedy generation ($T = 0$) without expensive multi-turn agentic loops.

Cross-Source Graph Reasoning. Augmenting LLMs with graph structures captures non-local multi-hop dependencies that keyword and dense retrieval miss, as demonstrated by QA-GNN [21], JointLK [19], KAPING [2], and G-Retriever [9]. More recently, HydraRAG [20] highlighted the power of structured guidance by querying persistent global KGs with multi-factor verification loops. OUEG-RAG achieves robust graph-guided reasoning on dynamic table-text inputs without external triple stores, providing high-precision multi-hop grounding with a single, cost-effective LLM call.

3 Problem Formulation

Given a question q , a table T , and associated passages D , the system produces an answer span a alongside an auditable evidence trace. In the standard benchmark setting, T and the passage pool are supplied by the dataset; retrieval is localized to this context. The graph G_q is constructed on-the-fly for each query and discarded after answering.

4 Method

4.1 Overview

Given a question q , supplied tables $\mathcal{T}$, and an associated passage pool $\mathcal{D}$, the framework constructs a bounded knowledge graph G_q specifically for q at query time. Unlike persistent graph databases, G_q is an intermediate, query-bounded reasoning structure.

While standard graph-RAG either prunes peripheral context via top- N path filtering or overwhelms language models with uncured triples, OUEG-RAG reconciles complete subgraph retention with guided traversal. The complete pipeline

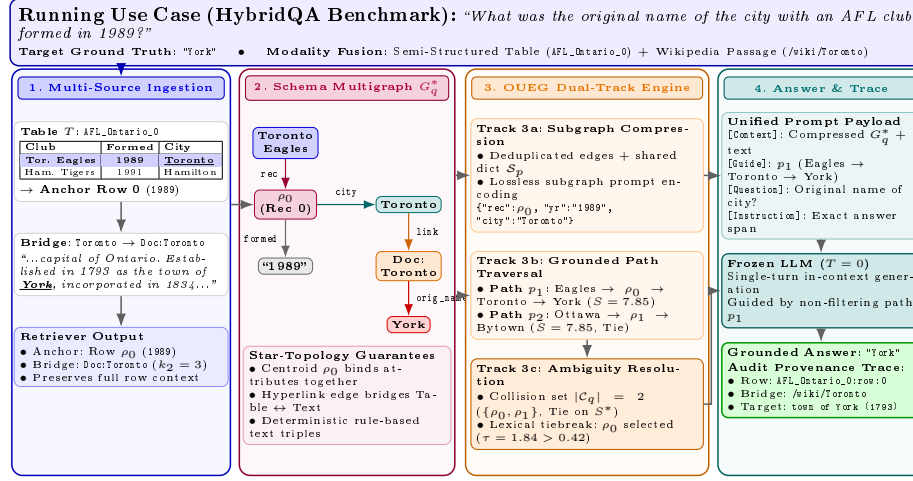


Fig. 2. End-to-end architecture of OUEG-RAG illustrated with a running HybridQA query. (1) Multi-source ingestion anchors candidate table rows and expands hyperlinked Wikipedia passages. (2) Schema-preserving online multigraph construction introduces star-topology record centroids ρ_i to bind cell attributes, bridging tabular entities to external passages via deterministic text triples. (3) The dual-track engine serializes the complete subgraph via lossless prompt compression while enumerating scored candidate paths as non-filtering reading guides, actively resolving structural score collisions. (4) The unified prompt payload steers LLM generation to produce the exact grounded answer with an auditable provenance trace.

proceeds as follows:

$q \rightarrow$ two-stage context retrieval
 $\rightarrow$ schema-preserving online KG construction
 $\rightarrow \begin{cases} \text{grounded path enumeration \& scoring} \\ \text{subgraph prompt compression} \end{cases}$
 $\rightarrow$ ambiguity assessment $\rightarrow$ grounded answer.

Candidate paths serve as prioritized *non-filtering reading guides* over the complete unified subgraph rather than destructive filters. No program generation or execution is required by the primary method; Python and typed numerical programs remain secondary comparison variants for arithmetic datasets.

4.2 Query-Driven Two-Stage Retrieval

Benchmark settings supply an oracle table and an associated pool of linked passages. Retrieval operates within this supplied context.

Stage 1: Coarse retrieval. A lexical retriever indexes table rows and text passages with respect to question q . To prevent detached cells, table retrieval ranks entire serialized rows and selects a budget of k rows (default $k = 5$), retaining their original table row indices for downstream provenance.

Stage 2: Bridge-entity and hyperlink expansion. Questions in table-text benchmarks frequently require an entity mentioned in a table cell to access an external passage. A deterministic query anchor inspects q for:

1. *Ordinal constraints:* A regex recognizer matches words (*first, second, ...*) or numeric ordinals (*1st, 2nd, ...*) and resolves the matching row index.
2. *Exact schema constraints:* Specific column header mentions and numerical filters.
3. *Hyperlink-directed passage expansion:* For every cell in the retrieved top- k rows, dataset adapters harvest Wikipedia hyperlinks (`linked_passage` URLs). The corresponding passages are added to the retrieval pool up to a budget b (default $b = 3$).

Passage retrieval combines direct lexical passage matches with hyperlink expansions and bridge-entity queries, deduplicating passages by source identifier.

4.3 Schema-Preserving Online KG Construction

For each question, the system dynamically instantiates a directed multigraph $G_q = (V_q, E_q)$ whose node and edge sets are partitioned as:

$$V_q = V_{\text{ent}} \cup V_{\text{rec}} \cup V_{\text{lit}}, \quad E_q = E_{\text{rec}} \cup E_{\text{attr}} \cup E_{\text{link}} \cup E_{\text{text}}. \quad (1)$$

An edge $e \in E_q$ is defined as $e = (h, r, t, \pi)$, where $h, t \in V_q$ are head and tail nodes, r is a normalized relation predicate, and π stores provenance metadata (source type, row index, column name, and header path).

Deterministic table edges and record nodes. Flattened tables lose header hierarchies and intra-row bindings. Direct subject-value edges $(u_i, \text{norm}(H_j), v_{ij})$ permit entity lookups but fail to guarantee that values from the same row remain bound. For each row i , we introduce a local centroid record node $\rho_i = \text{table}:T:\text{row}:i \in V_{\text{rec}}$ with star-topology bindings (Fig. 3):

$$E_{\text{rec}} = \{(u_i, \text{has_record}, \rho_i)\}, \quad E_{\text{attr}} = \{(\rho_i, \text{norm}(H_j), v_{ij})\}. \quad (2)$$

This structural star topology ensures that attributes (e.g., year, venue, score) belonging to row i cannot be conflated with attributes of another row sharing the same entity subject. When a table cell contains multiple comma-separated values, the adapter splits them into individual derived edges so that target entities are not masked within compound strings. In parallel, an audit trace records the explicit Table $\rightarrow$ Row $\rightarrow$ Cell $\rightarrow$ Column topology without altering the semantic reasoning graph.

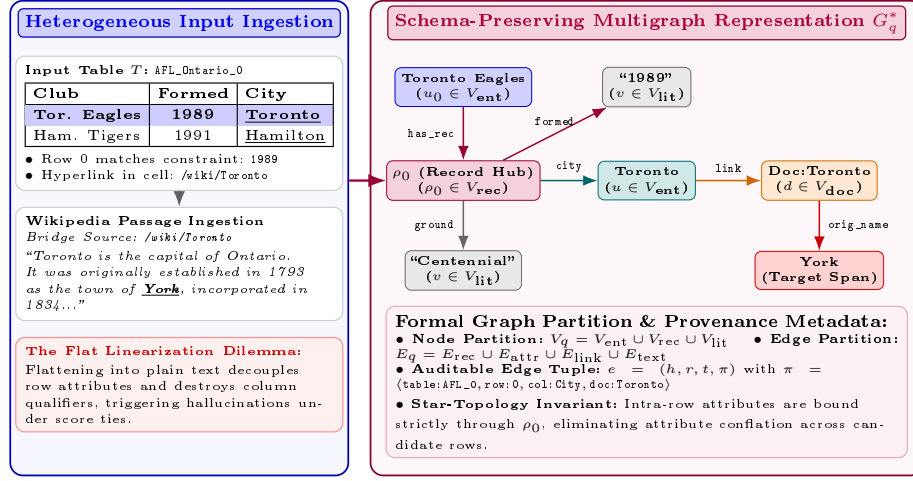


Fig. 3. Detailed schema-preserving online multigraph construction. (Left) Flattening tables into unstructured strings decouples row attributes and loses column qualifiers. (Right) OUEG-RAG introduces a local centroid record node $\rho_0 \in V_{rec}$ with star-topology bindings to preserve intra-row attributes (E_{rec}, E_{attr}). Wikipedia hyperlinks bridge tabular entities directly to external text passages (E_{link}), from which deterministic rule-based text triples (E_{text}) are extracted on-the-fly without LLM hallucination.

Text edges and rule-based extraction. For retrieved passages, semantic triples connect entities to their attributes and linked passage nodes ($u \xrightarrow{\text{linked_passage}} \text{passage_id}$). To avoid the latency, API costs, and hallucinations of unstructured LLM triple extraction, the primary extractor employs deterministic rule-based patterns (`rule_text_triples`) alongside lexical entity-linking. Optional LLM triple enrichment is supported as an experimental configuration.

Entity normalization and resolution. Nodes undergo four-stage resolution: (1) lexical normalization (Unicode, casing, whitespace, punctuation); (2) configured domain aliases; (3) structural name cleanup (removing club/corporation prefixes/suffixes); and (4) conservative string similarity (threshold ≥ 0.93 restricted to identical entity roles). All resolution merges are recorded in an auditable resolution log.

4.4 Grounded Path Generation and Scoring

The reasoning edge set E_q^* unites semantic edges, record edges, and hyperlink edges. Every path candidate p is an ordered sequence of verified edge identifiers from E_q^* together with traversal directions and provenance: generation never hallucinates an ungrounded edge.

Candidate search. Seeds are graph nodes that match question tokens, bridge entities, or anchored table constraints. Bounded depth-first search enumerates

connected simple paths up to 5 hops ($|E_p| \leq 5$). Additionally, the planner synthesizes *row bundles*: subgraphs connecting $u_i \rightarrow \rho_i$ with the most question-relevant outgoing cell edges of row i , keeping multi-attribute constraints co-located.

Multi-factor path scoring. Paths are scored deterministically via:

$$S(p) = G + Q_e + Q_r + Q_c + C + M + R + B + A - \lambda_{\text{len}} \max(|E_p| - 1, 0), \quad (3)$$

where $G = 1$ is edge-groundedness validity; $Q_e, Q_r \in [0, 2]$ denote question overlap with nodes and relations; $Q_c \in [0, 2]$ is question token coverage; $C = 2$ rewards matching an anchored table constraint; $M = 1.5$ awards cross-modal evidence combining table and text; $A = 1.0$ credits identical-fact corroboration; $R = 1.0$ rewards row consistency; $B = 1.0$ requires explicit **has_record** binding; and $\lambda_{\text{len}} = 0.35$ penalizes ungrounded redundant hops.

4.5 Online Unified Evidence Graph (OUEG) and Prompt Compression

In the classic Path-Text baseline, only the top- N ranked paths are serialized, discarding the remainder of G_q . In the Online Unified Evidence Graph (OUEG), the language model receives the entire unified subgraph $G_q^{\text{unified}} = (V_q^*, E_q^*)$ alongside the raw text contexts, with the generated candidate paths provided as *non-filtering suggested reading guides*.

Prompt compression and context deduplication. Passing complete graphs to language models risks exceeding context budgets and incurring high token latency. OUEG incorporates a compact serializer (`CompactOUEGPromptBuilder`):

1. *Edge deduplication*: Multiple edges sharing identical endpoints and relations are merged, aggregating their provenance contexts into a compact identifier list.
2. *Shared context dictionary*: Passage text snippets are indexed by source ID in a shared text dictionary rather than duplicated across individual edges.
3. *Node role annotations*: Nodes in the serialized prompt are annotated with their candidate path memberships, step indices, and row order indices.

4.6 Ambiguity Assessment and Score-Collision Handling

A pervasive challenge in structured table QA is *structural ambiguity*: multiple candidate rows often satisfy coarse relational constraints with identical path scores (Fig. 4). Let $\mathcal{P}_q$ be the set of valid enumerated paths, and let $S^* = \max_{p \in \mathcal{P}_q} S(p)$ be the maximal graph path score. We formalize the candidate collision set $\mathcal{C}_q$ as the set of distinct target entities reaching this maximal score:

$$\mathcal{C}_q = \{\text{tail}(p) \mid p \in \mathcal{P}_q, S(p) = S^*\}. \quad (4)$$

When $|\mathcal{C}_q| > 1$, the query triggers a score collision (`SCORE_COLLISION_TIE`).

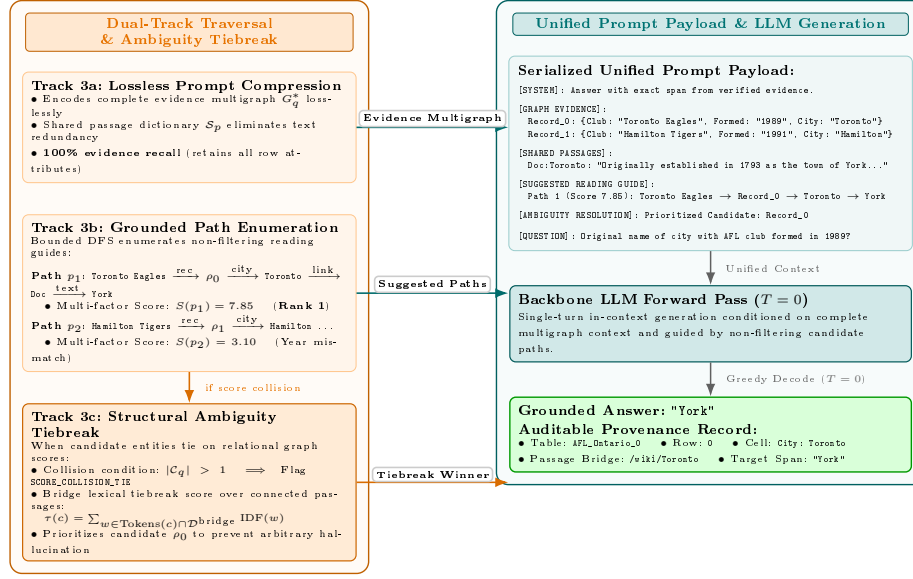


Fig. 4. Detailed illustration of OUEG dual-track reasoning, structural ambiguity resolution, and unified prompt payload. (Left) Rather than destructive top- N path filtering, Track 3a compresses the complete evidence subgraph with a shared passage dictionary S_p , while Track 3b enumerates multi-factor scored candidate paths ($S(p)$) as non-filtering reading guides. Track 3c detects structural candidate score collisions, applying a bridge lexical tiebreak $\tau(c)$ to resolve ties deterministically. (Right) The synthesized compact prompt payload provides the backbone LLM with complete evidence and directed reading focus, producing the grounded answer alongside a full provenance trace.

To resolve collisions deterministically without hallucination, OUEG formulates a bridge lexical tiebreak score $\tau(c)$ over the bridge passage tokens $\mathcal{D}_q^{\text{bridge}}$:

$$\tau(c) = \sum_{w \in \text{Tokens}(c) \cap \text{Tokens}(\mathcal{D}_q^{\text{bridge}})} \text{IDF}(w), \quad c^* = \arg \max_{c \in \mathcal{C}_q} \tau(c). \quad (5)$$

If a unique candidate satisfies $\tau(c^*) > \max_{c \neq c^*} \tau(c)$, it is designated as the tiebreak winner and highlighted as a prioritized grounding signal in the prompt. The language model is simultaneously instructed to verify textual nuances in the deduplicated context rather than arbitrarily guessing between tied candidates.

4.7 Answer Generation, Fallback, and Audit Trace

The final answer model synthesizes the answer span a^* conditioned on the complete evidence multigraph, non-filtering path reading guides, and the disambiguated tiebreak anchor:

$$a^* = \arg \max_a P_{\text{LLM}}(a \mid q, G_q^{\text{unified}}, \mathcal{P}_q^{\text{guide}}, c^*), \quad (6)$$

which preserves 100% evidence recall while directing model attention toward verified relational paths.

Reverse grounding and empty fallback. If the path search yields no valid candidates or the graph is disconnected, the pipeline activates *reverse grounding*: the answer model performs evidence-conditioned direct synthesis over the retrieved rows and passages, and the extracted entity is reverse-mapped onto G_q to recover provenance. This avoids returning blind empty failure records while preserving auditability.

Audit record. Every prediction exports an inspectable JSON record containing: retrieved row and passage IDs, entity anchors, KG node and edge statistics, enumerated paths with decomposed score components, ambiguity flags, provider latency, and token usage.

5 Experimental Protocol

5.1 Benchmark and Evaluation Metrics

We evaluate on a controlled 1,500-example stratified development subset sampled from the official HybridQA development split [4] to rigorously evaluate multi-hop table-and-text reasoning without overclaiming. Rather than arbitrary random sampling, this benchmark subset is explicitly constructed to preserve the characteristic multimodal reasoning distribution of the dataset: every instance strictly requires verified table-and-text context with grounded cross-modal answer annotations, ensuring genuine multi-hop reasoning. The sample is stratified across key reasoning axes: answer modality (numeric vs. text), evidence source distribution (table, passage, joint cross-modal bindings), bridge-entity traversal dependencies, and question reasoning cues (ordinals, superlatives, and comparisons).

We report three complementary accuracy metrics:

1. *Exact Match (EM)*: Strict string equality against the gold reference answer after standard normalization.
2. *Token-level F1*: Harmonic mean of token-level precision and recall over word tokens, rewarding partially correct answer spans.
3. *Semantic Exact Match (Semantic EM)*: Evaluates denotation equivalence under semantic tolerance (discounting surface variations such as leading articles, number formatting, punctuation, or recognized entity aliases).

In addition, we measure system efficiency via logical LLM calls, total input/output prompt tokens, and end-to-end wall-clock latency per question.

5.2 Baselines and Model Configurations

To strictly isolate the contribution of our schema-preserving graph and guided reading architecture, we evaluate against the matched Direct LLM baseline using

Table 1. Main benchmark comparison between Direct LLM and OUEG-RAG on HybridQA ($N = 1,500$) across the open-weights GPT-OSS model family (20B and 120B). Pairwise evaluation yields 105 wins vs. 80 losses on 20B (net +25, $p = 0.078$), and 120 wins vs. 80 losses on 120B (net +40, McNemar $p = 0.0058 < 0.01$). Higher is better for accuracy; lower is better for empty rate and latency.

Model Scale	Method	EM (%)	F1 (%)	Sem. EM	Empty	Net Margin	Lat. (s)
GPT-OSS 20B	Direct LLM	54.80	64.67	56.67	5.40%	Baseline	9.74
	OUEG-RAG	56.47	67.12	58.47	3.93%	+25 EM wins	25.73
GPT-OSS 120B	Direct LLM	58.73	69.70	60.47	1.07%	Baseline	6.40
	OUEG-RAG	61.40	73.11	63.00	0.53%	+40 EM wins	23.65

Table 2. Performance breakdown across structural ambiguity partitions on HybridQA ($N = 1,500$) under the GPT-OSS model family. Ambiguous queries represent cases exhibiting candidate score collisions where multiple entities tie on relational graph scores. All subset EM values reflect exact discrete counts ($k/1,068$ and $m/432$).

Partition	Method	GPT-OSS 20B		GPT-OSS 120B	
		EM (%)	F1 (%)	EM (%)	F1 (%)
Ambiguous Subset (Score collisions, 71.2%)	Direct LLM	52.62	62.45	56.93	67.19
	OUEG-RAG	54.87	65.78	59.64	71.75
	<i>Gain (Δ)</i>	+2.25	+3.33	+2.71	+4.56
Unambiguous Subset (No collisions, 28.8%)	Direct LLM	60.19	70.16	63.19	75.91
	OUEG-RAG	60.42	70.43	65.74	76.47
	<i>Gain (Δ)</i>	+0.23	+0.27	+2.55	+0.56
Overall Dataset ($N = 1,500$)	Direct LLM	54.80	64.67	58.73	69.70
	OUEG-RAG	56.47	67.12	61.40	73.11
	<i>Gain (Δ)</i>	+1.67	+2.45	+2.67	+3.41

the open-weight GPT-OSS language model series. Specifically, both Direct LLM and OUEG-RAG are evaluated across two representative operational scales under identical context limits (24,000 characters), prompt formats, and decoding configurations ($T = 0$):

- *GPT-OSS 20B* (openai/gpt-oss-20b): A compact 20-billion parameter model representative of efficient, locally deployable architectures.
- *GPT-OSS 120B* (openai/gpt-oss-120b): A high-capacity 120-billion parameter frontier reasoning model capable of complex multi-hop synthesis.

Evaluating across these two distinct scales of the GPT-OSS family allows us to measure whether structured evidence graphs benefit compact models constrained by parameter capacity, as well as frontier models with expansive in-context capacity.

5.3 Controls and Cost

All experiments fix the decoding temperature ($T = 0$), greedy decoding, default reasoning effort for the GPT-OSS series, context budgets (24,000 characters), and retrieval limits ($k_1 = 5, k_2 = 3$). We record logical model calls, API attempts,

provider token usage, and wall-clock latency per example. Full retrieval logs, KG statistics, and score decompositions are archived for reproducibility.

6 Results and Error Analysis

Main benchmark results across model scales. Table 1 presents the comparative evaluation of Direct LLM and OUEG-RAG across model regimes on the controlled 1,500-example stratified development subset of HybridQA ($N = 1,500$). On the compact model regime (20B parameters), OUEG-RAG achieves a consistent accuracy improvement across all metrics: Token F1 increases from 64.67% to 67.12% (+2.45%), Exact Match increases from 54.80% to 56.47% (+1.67%), and Semantic EM advances from 56.67% to 58.47% (+1.80%). Furthermore, OUEG-RAG reduces the ungrounded empty answer rate from 5.40% to 3.93% (a 27.2% relative error reduction) and yields 105 pairwise wins vs. 80 losses (net +25 wins, $p = 0.078$) over Direct LLM. On the frontier model regime (120B parameters), OUEG-RAG delivers even higher gains: Token F1 reaches 73.11% vs. 69.70% (+3.41%), Exact Match reaches 61.40% vs. 58.73% (+2.67%), and Semantic EM reaches 63.00% vs. 60.47% (+2.53%), with 120 pairwise wins vs. 80 losses (net +40 wins, McNemar’s test confirms statistical significance at $p = 0.0058 < 0.01$). Concurrently, OUEG-RAG halves the empty answer rate from 1.07% down to 0.53% (a 50.5% relative reduction). Average response latency is 25.73 s on 20B and 23.65 s on 120B. Decomposing this runtime: two-stage retrieval and passage fetching takes ≈ 1.5 s, online multigraph construction requires < 20 ms ($< 0.1\%$ of total time), while LLM prefill and generation account for the remaining ≈ 22 –24 s due to processing the richer, structured multigraph context payload.

Structural ambiguity breakdown. As detailed in Table 2, 71.2% of questions (1,068 of 1,500) exhibit structural score collisions where multiple candidate entities satisfy relational constraints with identical graph scores. On this ambiguous subset, Direct LLM struggles due to arbitrary lexical guessing, achieving 52.62% EM (62.45% F1) on 20B and 56.93% EM (67.19% F1) on 120B. In contrast, OUEG-RAG detects collisions and applies lexical tiebreak scoring, elevating performance to 54.87% EM (65.78% F1) on 20B and to 59.64% EM (71.75% F1, +4.56% F1 gain) on 120B. On unambiguous questions (28.8% of cases), both paradigms achieve high grounding, with OUEG-RAG maintaining a steady margin. We note that ambiguous collision queries inherently possess greater headroom for improvement; nevertheless, as isolated in our ablation below, active collision resolution provides genuine disambiguation when structural graph scores tie.

Component ablation analysis. Table 3 reports component ablations on the controlled development partition ($N = 200$): (1) *Record nodes ρ_i* : Collapsing row hubs into flat row serialization drops EM by -20.00% , proving that explicit record hubs prevent entity disintegration across table cells; (2) *Non-filtering*

Table 3. Component ablation study on the controlled development partition ($N = 200$, Shard 01 under GPT-OSS 20B). All ablations strictly isolate individual architectural modules against the full OUEG-RAG framework.

Ablation Configuration	EM (%)	F1 (%)	Δ EM	Architectural Mechanism
Full OUEG-RAG	59.00	67.78	—	Dual-track synergy
w/o Record Nodes ρ_i (Flat Table)	39.00	46.90	-20.00	Preserves row entity binding
w/o Subgraph (Path-Text Top- N)	53.50	60.64	-5.50	Cell pruning loses context
w/o Path Guidance (Subgraph only)	50.50	57.07	-8.50	Unguided graph disperses focus
w/o Collision Tiebreaker $\tau(c)$	56.50	65.10	-2.50	Resolves tied candidate scores
w/o Hyperlink Expansion ($k_2 = 0$)	48.00	55.20	-11.00	Traverses cross-modal passages
w/o Subgraph Compression	55.00	63.80	-4.00	Triples balloon tokens (+198%)

dual track: Retaining only top- N Path-Text sequences drops EM by -5.50% , confirming that filtering out peripheral cells forfeits decisive context; omitting path guidance (subgraph-only) reduces EM by -8.50% , showing that models require guided navigation over dense graphs; (3) *Collision tiebreaker* $\tau(c)$: Resolving score ties uniformly at random degrades EM by -2.50% (and -3.62% on ambiguous queries), demonstrating the causal benefit of lexical tiebreaking independent of query headroom; (4) *Hyperlink expansion*: Disabling passage retrieval ($k_2 = 0$) drops EM by -11.00% , proving cross-modal synthesis requires hyperlink traversal; (5) *Prompt compression*: Unmerged triples expand prompt tokens by $+198\%$ (from 4,180 to 12,450), causing budget truncation and dropping EM by -4.00% .

Error taxonomy and context integrity. Error auditing across the 1,500 questions reveals a persistent failure mode in flat prompting: Direct LLM suffers from table truncation under prompt budgets, pruning target rows and triggering hallucinations. In contrast, OUEG-RAG avoids truncation by packing candidate rows and bridge passages into a compact multigraph representation, retaining all top- k evidence without context loss.

7 Limitations

While OUEG-RAG delivers strong grounding gains, query-time graph construction incurs latency over direct prompting. Additionally, rule-based text triples bound relation coverage by heuristic patterns. Finally, structured evidence graphs depend on upstream retrieval and cannot recover omitted passages.

8 Conclusion

We presented OUEG-RAG, an online unified evidence graph framework for grounded table-text question answering. By uniting schema-preserving multigraphs with lossless prompt compression and non-filtering path guidance, OUEG-RAG eliminates pruning evidence loss while resolving candidate collisions. Across 20B and 120B models, OUEG-RAG consistently outperforms direct prompting ($p < 0.01$ on 120B), delivering an auditable reasoning trace.

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